BANTAY–BAHA

Early warning for flood-linked disease: which provinces will see a surge after a typhoon, and which of them will be cut off from a hospital.

46%
lower forecast error than the seasonal baseline

7 / 10
top-ranked provinces surged after Typhoon Ompong

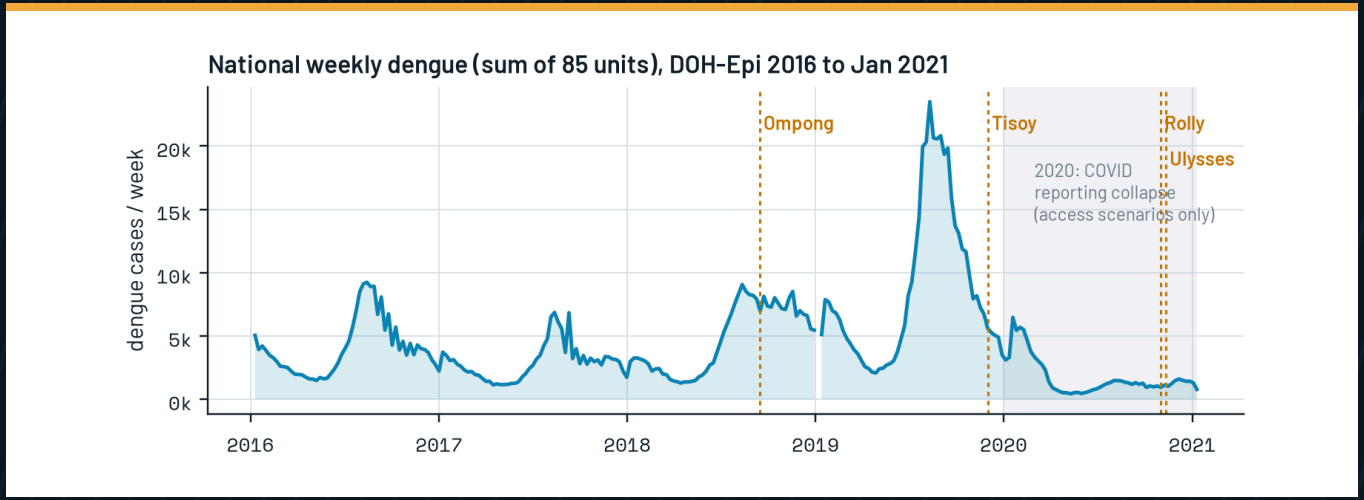
100
ferry routes a single typhoon can close, out of 108

TYPHOONS BRING FLOODS. FLOODS BRING DISEASE.

8–9
tropical cyclones cross the Philippines every year, mostly from July to October (PAGASA)

2,089
leptospirosis cases and 162 deaths after Typhoon Ondoy in 2009, five times the national lepto deaths of 2008 (Balbuena et al., 2010)

146k
dengue cases from January to July 2019, when DOH declared a national epidemic



WHO IS HIT HARDEST

Floodwater spreads leptospirosis within days; standing water breeds the mosquitoes behind dengue for weeks after. The same storm closes roads and grounds ferries, so the most exposed communities are often the ones that **cannot reach a hospital**.

THE RESPONSE ARRIVES AFTER THE SURGE

A typhoon sets off a sequence. Each step is predictable from data that already exist, yet today's response starts only at the last one.

DAY 0
LANDFALL
Gale-force winds; ferries suspended; roads blocked by landslides and debris

DAYS 1-7
FLOODING
Rivers carry upstream rain to the lowlands; towns at river mouths flood after the rain stops

WEEKS 1-2
LEPTOSPIROSIS
Risk from rainfall peaks about two weeks later and runs through flooding (Matsushita et al., 2018)

WEEKS 2-8
DENGUE
Breeding sites multiply; cases rise province by province

TODAY
RESPONSE
Prophylaxis, vector control and medical teams go out after cases rise; isolated places are reached last

What is missing is a tool that looks at **this** storm and says, before the roads close, **where** disease will rise and **who** will be cut off.

DENGUE FORECASTS EXIST
Climate-driven, province-level (Buczak et al., 2014), but not for a specific storm and without honest ranges

ACCESS STUDIES EXIST
Flood access loss for one city, Metro Manila (Tariverdi et al., 2023), with no health forecast and no ferries

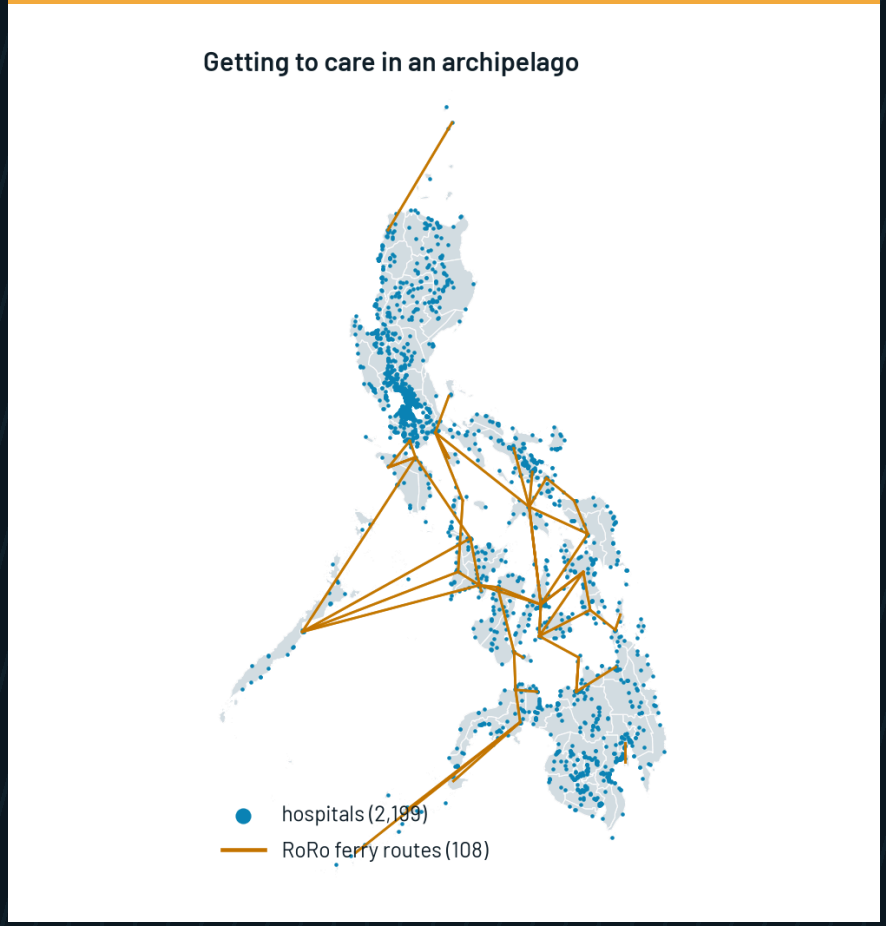
OUTBREAK WAVES ARE KNOWN
Dengue travels between regions (van Panhuis et al., 2015), but this is unused for storm response

AN ARCHIPELAGO MAKES ACCESS THE PROBLEM

- **An island nation.** Many provinces reach the nearest hospital only by RoRo ferry; a gale warning cancels the crossing
- **Roads in floodplains.** The road to the provincial hospital often runs exactly where the river overflows
- **Measured once, for one city.** A 15 cm flood leaves about 22% of Metro Manila without access to higher-level care (Tariverdi et al., 2023); nothing comparable exists nationwide, or with ferries

GUIDING QUESTION

Can flood exposure, a graph of outbreak spread and a storm-disrupted transport network, combined with calibrated forecasts, rank the provinces that need help first, early enough to act?



THREE GRAPHS, ONE RANKING

1 / FLOOD EXPOSURE

Sentinel-1 observed flood, CHIRPS + IMERG rain, rain routed down rivers, UP NOAH hazard, typhoon gale swaths

2 / LEAD-LAG GRAPH

PCMCI+ finds which provinces rise first; their recent rise feeds every follower's forecast

3 / ACCESS GRAPH

OSM roads + RoRo ferries + hospitals, and a friction raster as a second method; a storm cuts both

QUANTILE FORECAST

XGBoost, 23 quantiles, 1-4 weeks ahead, tested by rolling origin against zero-shot foundation models and a graph network

A / ACCESS LOSS

share of people who lose their route to a hospital, with the range between methods

R / SURGE RISK

calibrated chance the next 4 weeks beat the province's usual bad month

P = R × A

ranked provinces, drivers and an action tier for a real or shifted track

Each half is validated on its own: forecast vs a seasonal baseline and benchmarks; access vs Metro Manila and Copernicus.

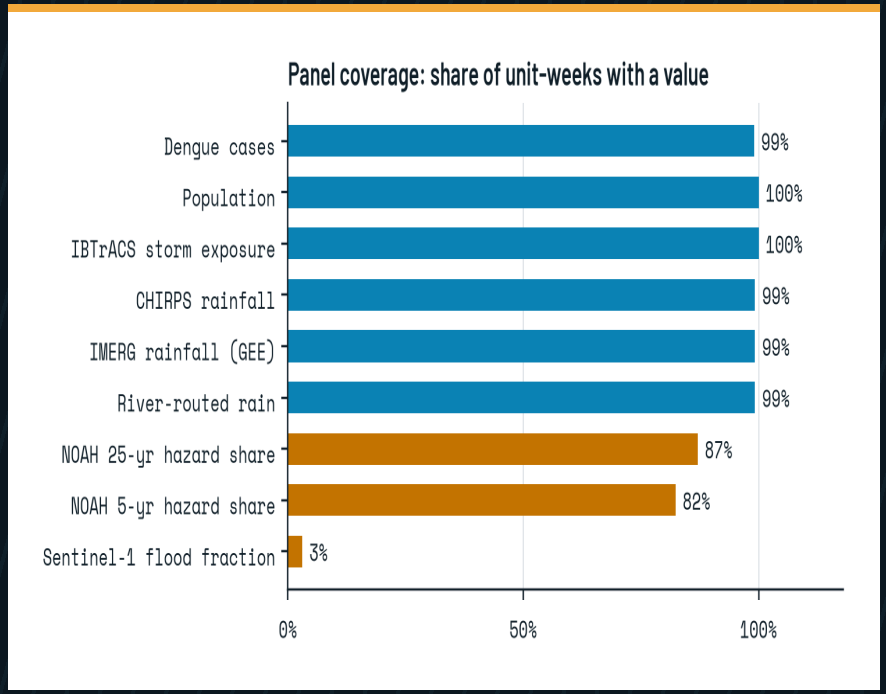
501 OPEN FILES, 22 SOURCES

LAYER	SOURCES
Disease	DOH-Epi dengue, province/city x week, 2016-2021 (HDX); Project CCHAIN city leptospirosis; national PIDSR; OpenDengue
People and places	2020 census; COD-AB boundaries; WorldPop 2020 100 m; Meta HRSL under-5 / 60+
Rain and rivers	CHIRPS v2.0; GPM IMERG V07; HydroRIVERS
Storms and flood	IBTrACS; Sentinel-1 radar (Google Earth Engine); UP NOAH hazard maps; Copernicus EMS
Getting to care	OpenStreetMap roads; MARINA RoRo routes (roroph); OSM and NOAH health facilities; MAP friction surface

22,270
province-weeks (85 x 262)

2,199
hospitals and 108 ferry routes in the access graph

369k
1 km population cells, 109 M people



Only aggregated, non-identifiable health data.

THE DATA FOUGHT BACK. HOW WE HANDLED IT

WHAT WE FOUND	WHAT WE DID
126 dengue reporting units, not 81 provinces: cities mixed in, misspellings, boundaries changed since 2020	A reviewed crosswalk to 85 harmonised 2016-2020 provinces; a week counts only if every member reported
2020 reporting collapsed under COVID-19 (91k vs 442k cases)	2020 is never a forecast target; it is used for access scenarios only
Flood-hazard maps missing for 11 provinces	Carried as missing, never as zero hazard
Satellite flood maps disagreed with Copernicus at first	Same-orbit change detection with a dB-difference rule, validated against Copernicus EMS
Short series make causal-discovery algorithms over-confident	False discovery rate control, tested on pure noise first
Surge probability was under-confident in the middle range	Isotonic recalibration, fitted only on forecasts whose outcome was already known
Road network and friction raster disagree once a storm hits	Both are run on every track; the gap is shown as a range, not hidden in one number

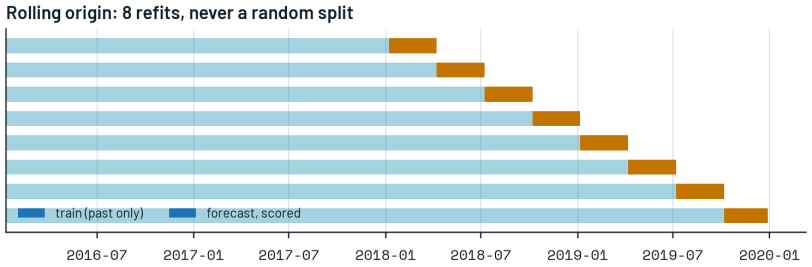
HONEST RANGES, TESTED LIKE REAL TIME

MODEL

- One pooled XGBoost per horizon, 1-4 weeks ahead
- 23 quantiles (the CDC FluSight standard), so every forecast is a range
- Inputs: recent cases, season, rain, river, storm, flood, leader provinces
- A direct 4-week model gives the surge probability R, recalibrated on past forecasts only; SHAP explains it

EVALUATION WITHOUT LEAKAGE

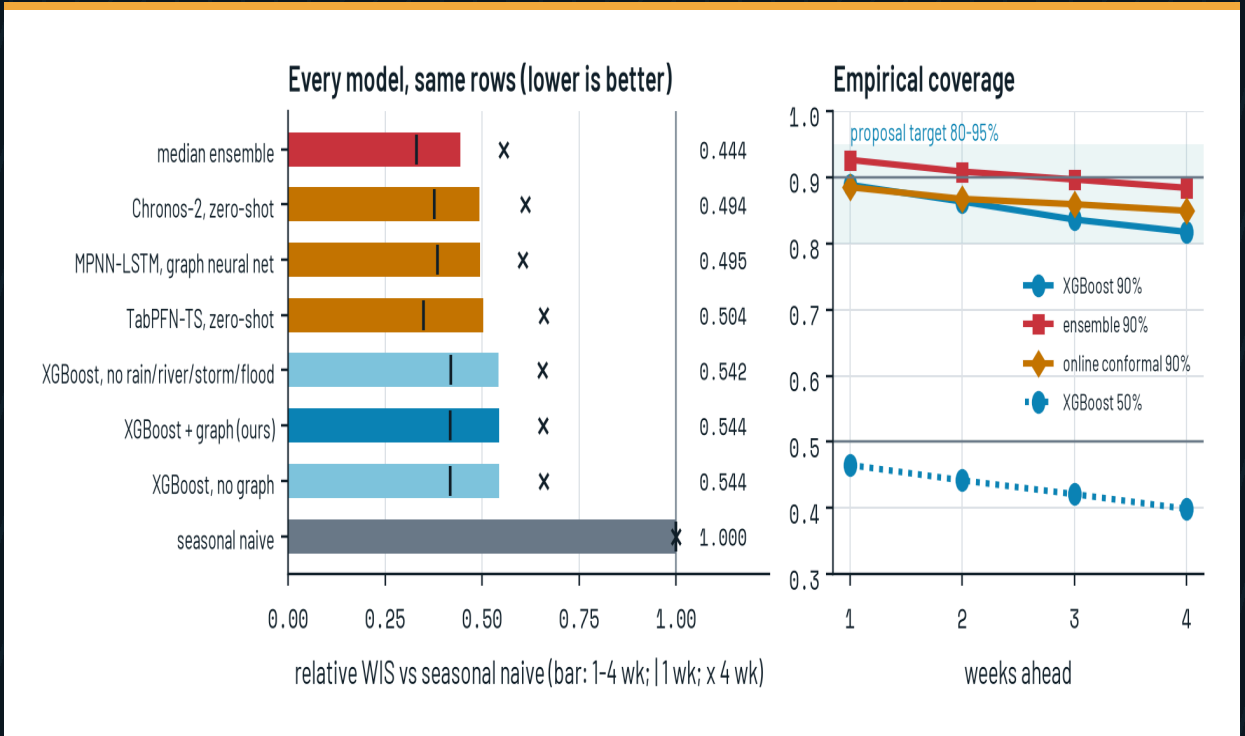
- Rolling origin over 2018-2019: refit every 13 weeks on past data only
- Graph, thresholds and calibration are re-learned inside each window
- Weighted interval score on case counts (Bracher et al., 2021) vs "same week last year"
- Benchmarks on the same rows: Chronos-2, TabPFN-TS, MPNN-LSTM, and their median ensemble



34,050

out-of-sample forecasts scored: 85 provinces, 4 horizons, 2018-2019

BEATS THE BAR BY 46%; AN ENSEMBLE BY 56%



MODEL	REL WIS	VS XGB	MEDIAN WIS	COV 90
median ensemble	**0.444**	**0.816**	**6.29**	**0.90**
Chronos-2, zero-shot	0.494	0.908	6.54	0.88
MPNN-LSTM, graph neural net	0.495	0.910	6.83	0.88
TabPFN-TS, zero-shot	0.504	0.926	6.72	0.87
XGBoost, no rain/river/storm/flood	0.542	0.997	6.26	0.85
XGBoost + graph (ours)	**0.544**	**1.000**	**6.23**	**0.85**
XGBoost, no graph	0.544	1.001	6.24	0.85
seasonal naive	1.000	1.839	14.97	0.85

- The **ensemble** is best on average (0.44); the benchmarks win on epidemic peaks
- On the **typical** province-week XGBoost is best (median WIS 6.23 vs 6.54-6.83)
- Storm scenarios use XGBoost: the only model that reads the storm track

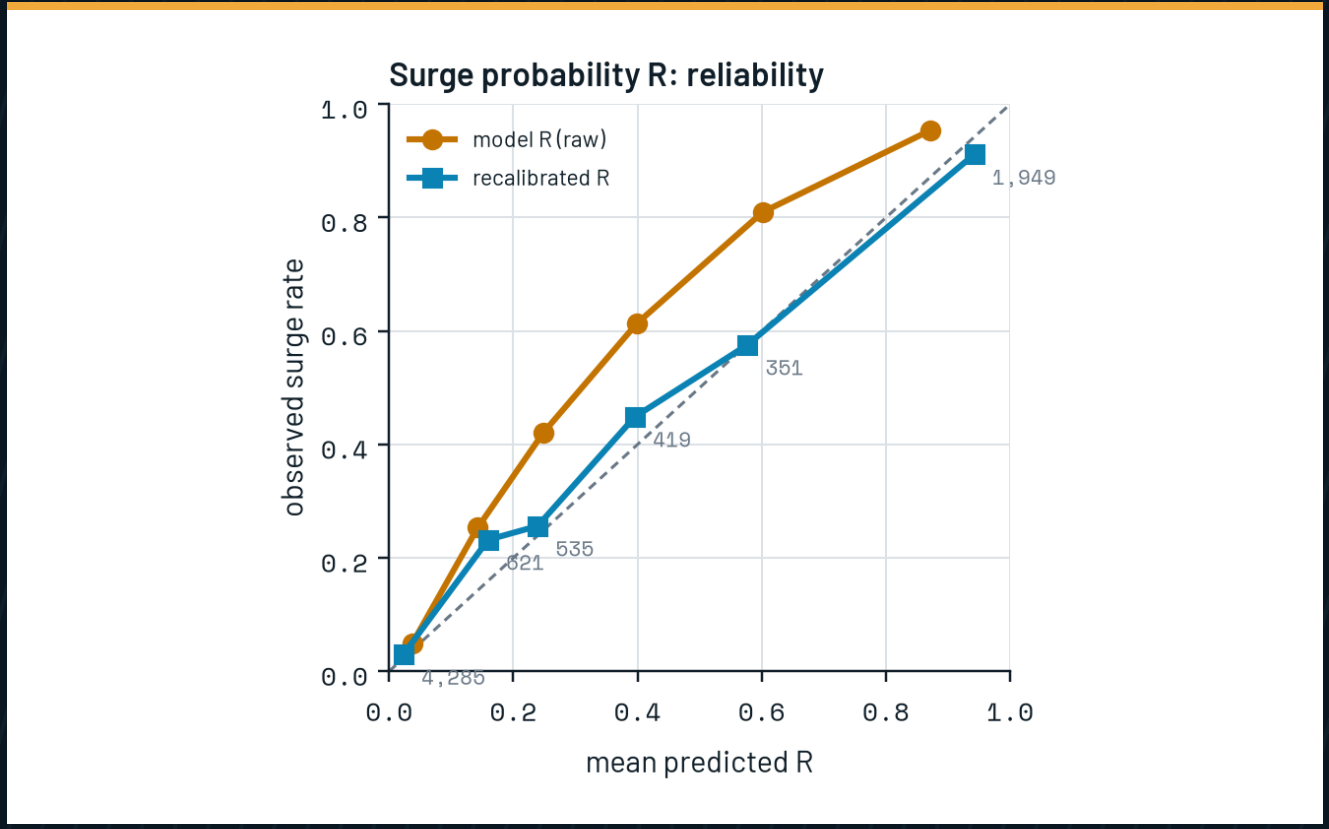
34,050 forecasts, every model on the same rows; XGBoost 90% coverage 85% (target 80-95%).

SURGE RISK R, CALIBRATED ON THE PAST

0.089
Brier score of the model's R

0.082
after recalibration (0.216 if always predicting the base rate)

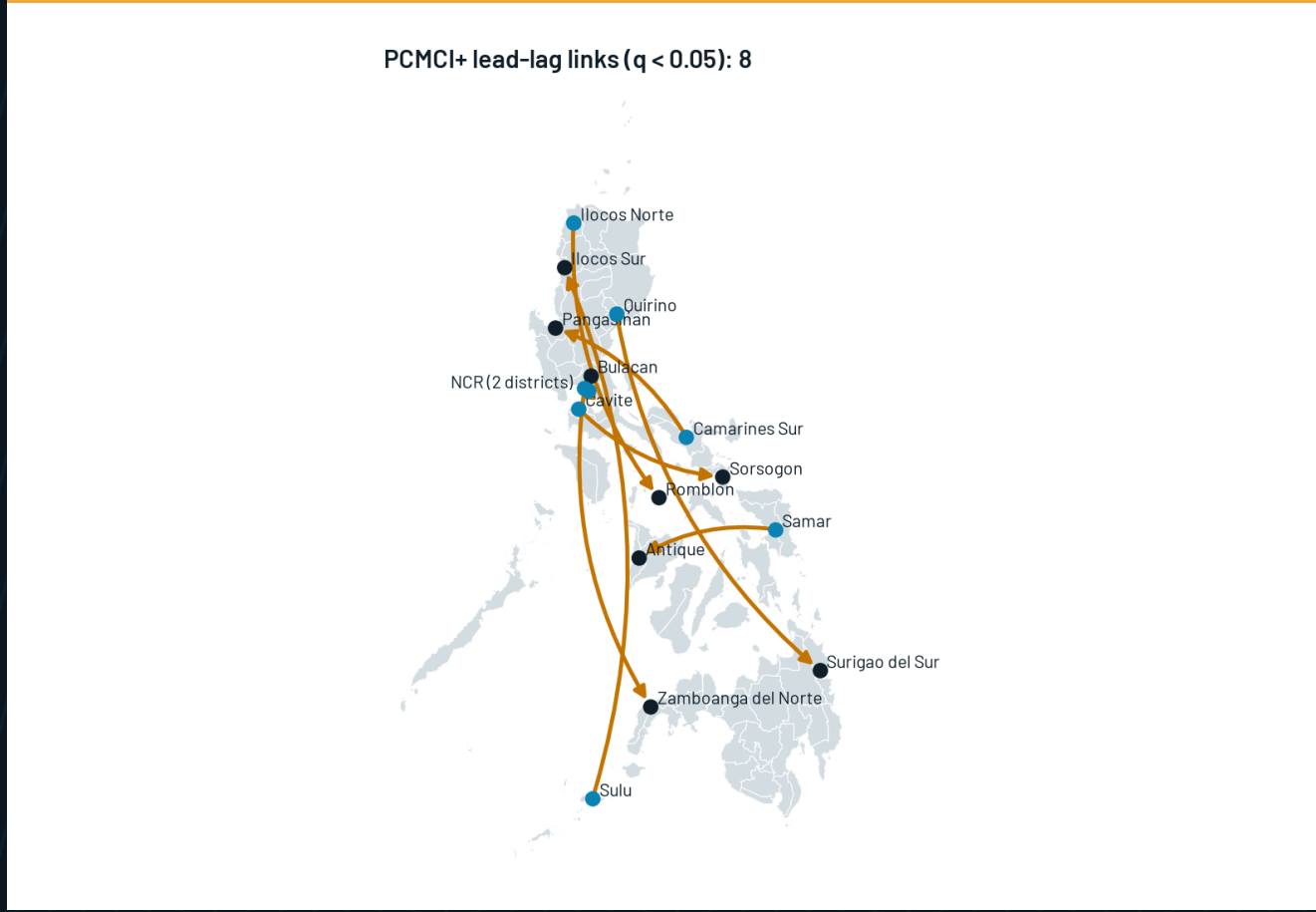
- The model under-stated mid-range risk; an isotonic map fitted only on **past** forecasts with known outcomes fixes it
- When R says high, surges follow: above 0.7, 91% of province-weeks surged
- 2019's epidemic made surges common (32% of province-weeks)



HOW OUTBREAKS TRAVEL BETWEEN PROVINCES

- PCMCI+ separates "A rises, then B" from "both rise in the rainy season" (Runge, 2020)
- **8 lead-lag links** survive false-discovery control; plain correlation would have drawn 4,265
- Leaders are marked as **sentinel sites** on the dashboard: places to watch first

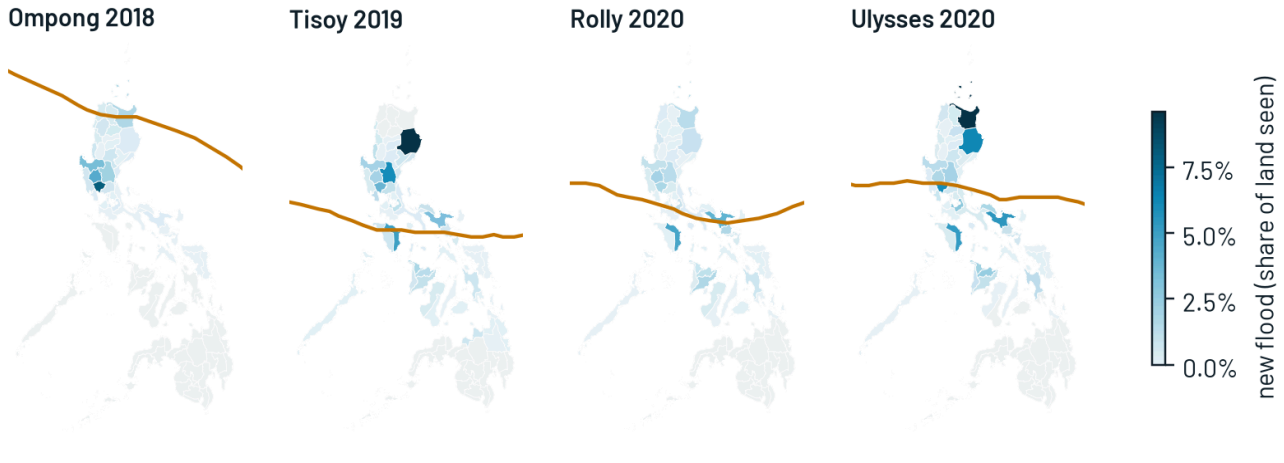
LEADER	FOLLOWER	LEAD	BOOTSTRAP
Cavite	Sorsogon	1 wk	70%
NCR Second	Bulacan	3 wk	70%
Quirino	Surigao del Sur	1 wk	65%
Ilocos Norte	Romblon	1 wk	75%
NCR Third	Zamboanga del Norte	1 wk	85%



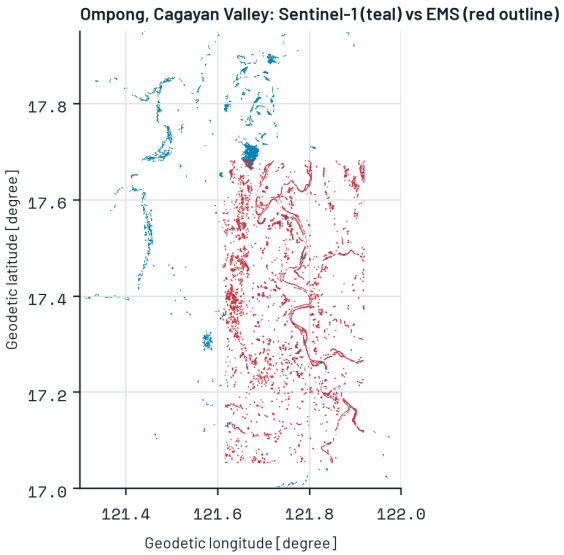
SIGNAL 12 / RESULT 4

DATA: OBSERVED FLOOD EXPOSURE

FLOOD SEEN FROM SPACE, TOWN BY TOWN

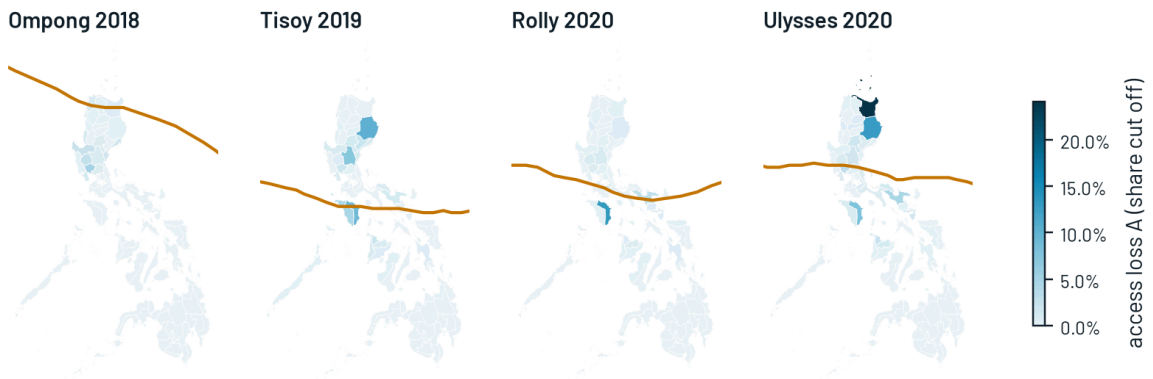


TYPHOON	PEOPLE IN NEW FLOOD	MUNICIPALITIES >1% FLOODED	MOST PEOPLE FLOODED
Ompong	0.5 M	131	Candaba, Pampanga
Tisoy	1.1 M	216	Jaen, Nueva Ecija
Rolly	0.6 M	173	Naujan, Oriental Mindoro
Ulysses	1.4 M	249	Tuguegarao City, Cagayan

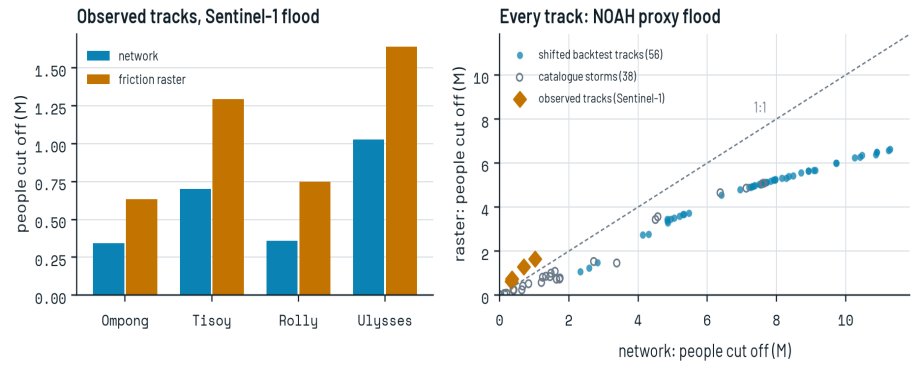


Checked against Copernicus EMS (Ompong): 55% of Sentinel-1 flood inside the EMS areas matches EMS polygons; Sentinel-1 recovers 24% of the EMS-mapped area (one radar pass every 6-12 days misses short floods).

WHO GETS CUT OFF: TWO METHODS, FOUR TYPHOONS

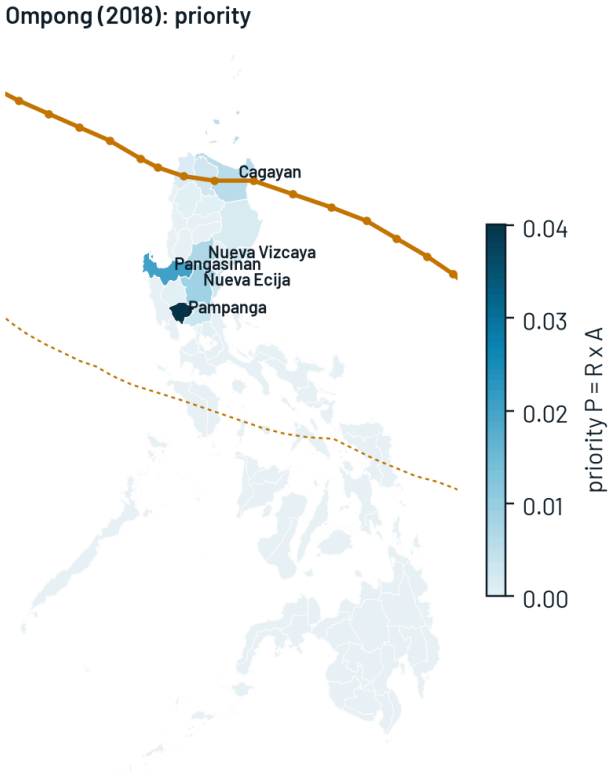


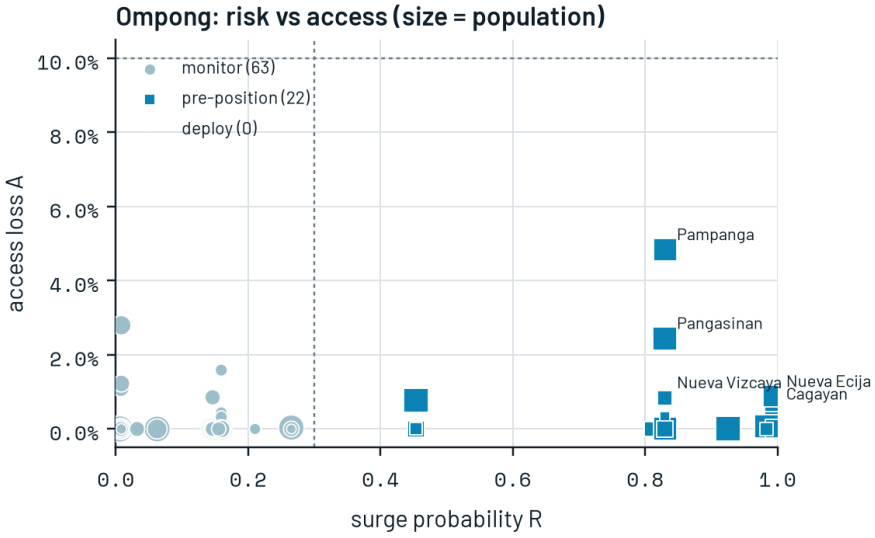
TYPHOON	FERRIES CUT	CUT OFF: NETWORK	RASTER	UNDER-5	60+
Ompong	42	0.3 M	0.6 M	27k	18k
Tisoy	100	0.7 M	1.3 M	65k	41k
Rolly	42	0.4 M	0.7 M	34k	22k
Ulysses	54	1.0 M	1.6 M	96k	63k



- Road network and friction raster agree on a normal day (ρ 0.95 across provinces) but not on the size of a storm's cut: the range is the uncertainty
- Radar flood: a **lower bound**. Hazard maps on the shifted tracks: 2.3 M–11.3 M, an **upper bound**
- Older people are under-counted in the population map (6.7% vs 8.5% in the census)

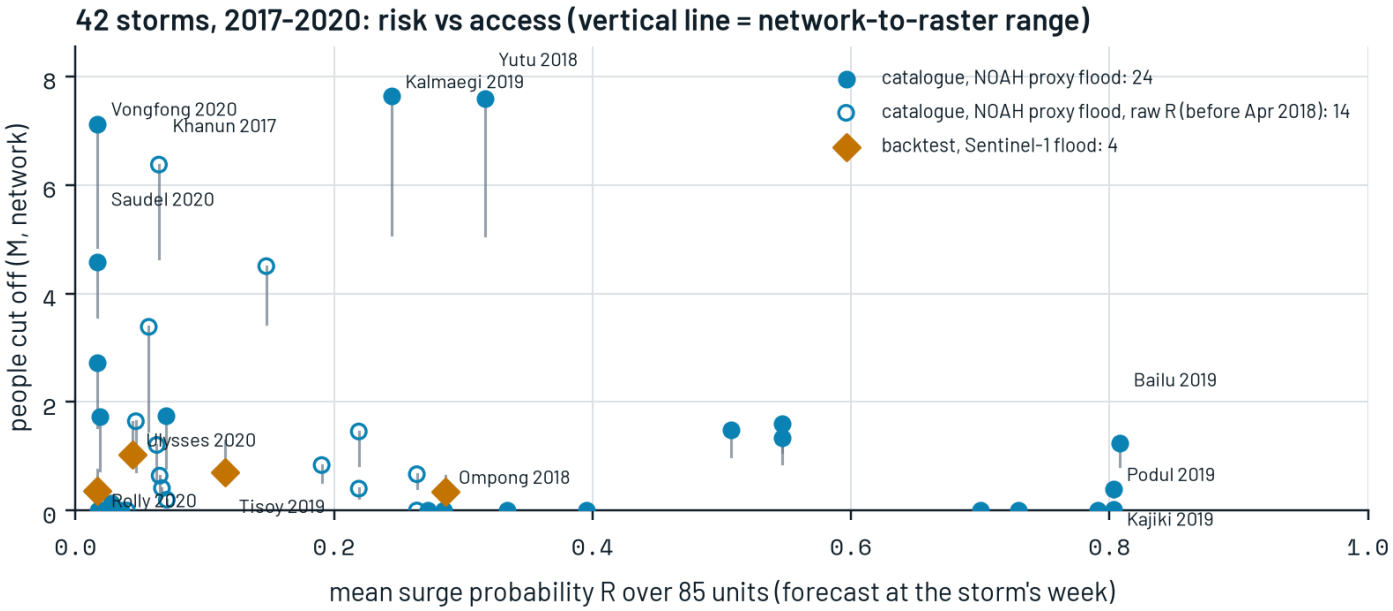
OMPONG 2018: PAMPANGA FIRST





Using data up to the landfall week only: 22 provinces flagged to **pre-position supplies** before the surge. In the next four weeks, **7 of the top 10** really surged.

EVERY STORM 2017–2020, NOT JUST FOUR



42

storms scored: 38 more beyond the four backtests, each with the model of its own week

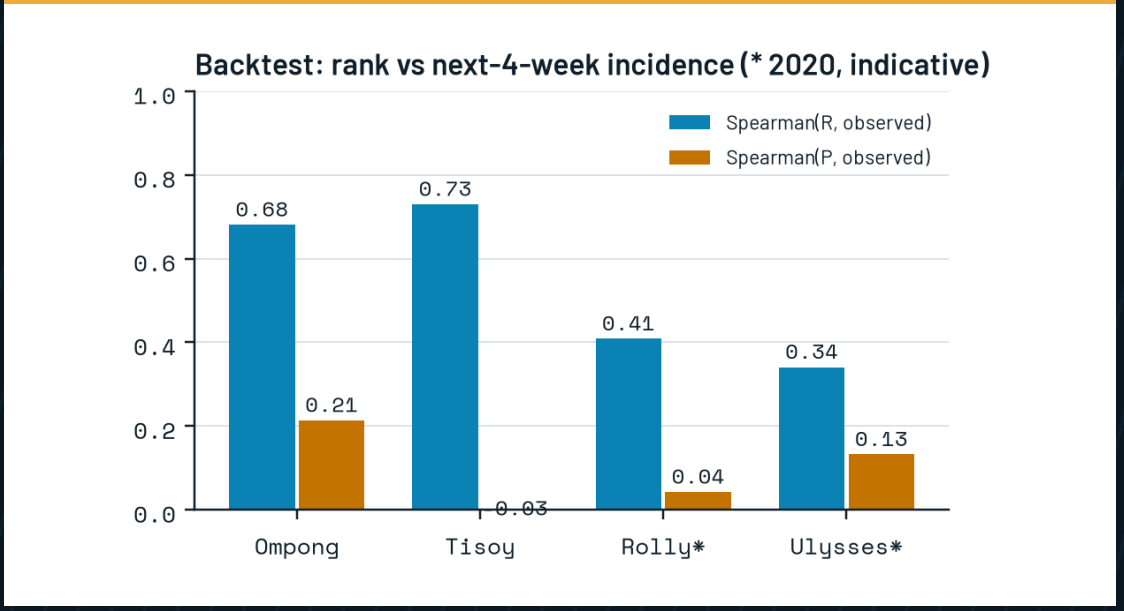
7.6 M

most people cut off: Kalmaegi 2019, flood from hazard maps (an upper bound)

0.81

highest mean surge risk: Bailu 2019, in the 2019 epidemic season

IT POINTS AT THE RIGHT PLACES



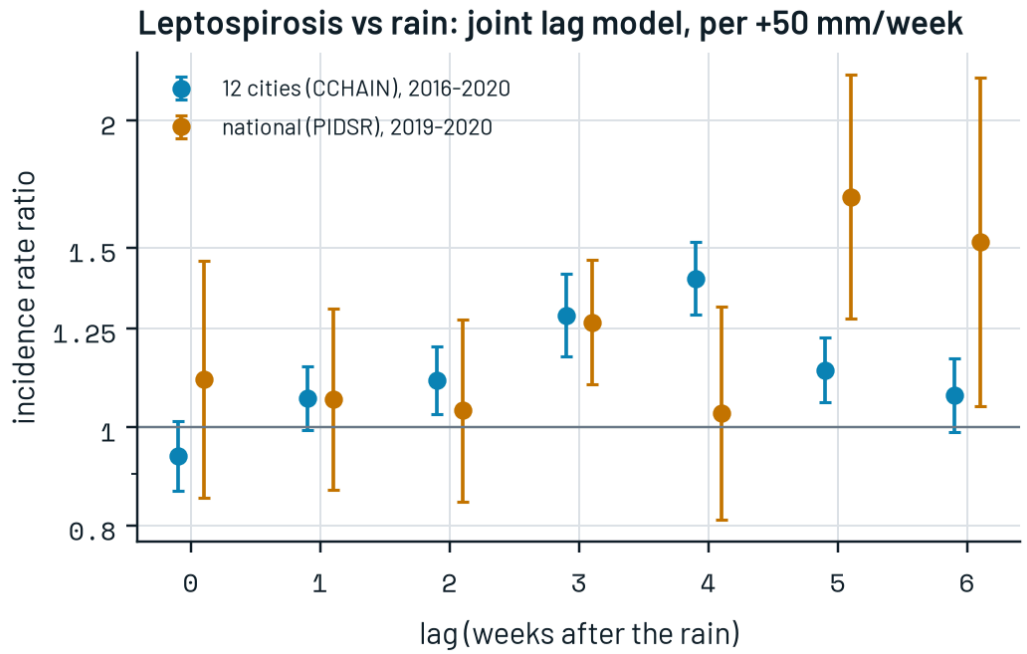
- **Backtest**: R's ranking matches the next four weeks' incidence, ρ = 0.68 (Ompong) and 0.73 (Tisoy)
- **Ompong**: 7 of the 10 highest-priority provinces surged, against a 40% base rate
- **Robust**: nine changes to travel speeds, ferry rules, cut-off rules and storm radius leave the ranking in place on both access methods (Kendall's τ ≥ 0.99)
- **Benchmark**: 30% of Metro Manila cut off when hazard-zone roads close; Tariverdi et al. (2023) found ~22% for a 15 cm flood, the same order

0.73
rank correlation, Tisoy

≧0.99
ranking stability (Kendall's τ)

30%
NCR cut off vs ~22% published

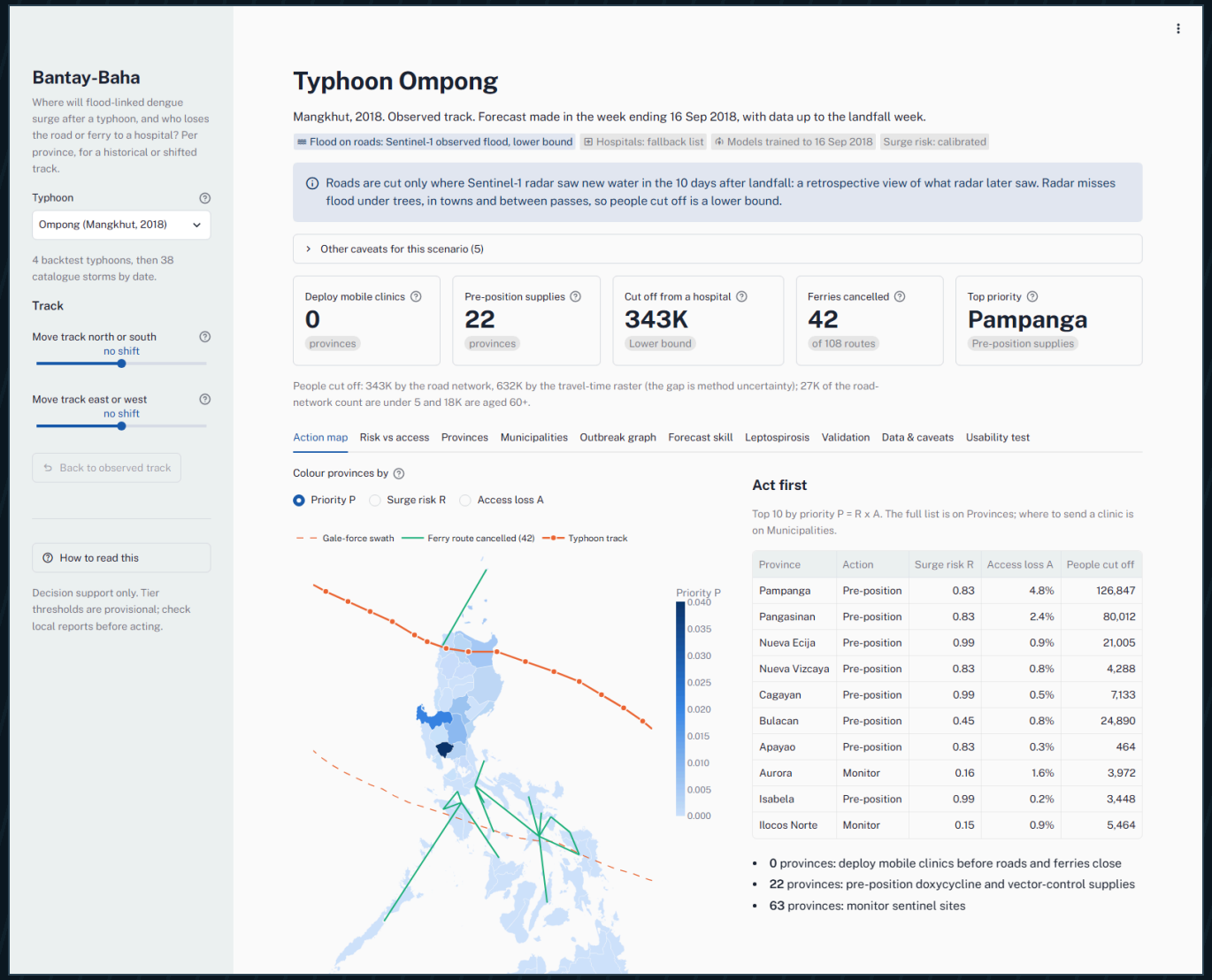
LEPTOSPIROSIS: RAIN NOW, CASES IN A MONTH



1.40×

leptospirosis cases 4 weeks after an extra 50 mm of weekly rain, in 12 cities (1.29–1.52)

- Nationally the same pattern, but mostly from 2020
- After Rolly and Ulysses, cities in the gale swath had 4.3× the expected cases; not after Ompong or Tisoy
- An association in 12 cities: not extrapolated to provinces, not yet in the ranking



WHAT A HEALTH OFFICER SEES

- Pick a typhoon, or move its track 0.5-1° south or north, or 1° east or west
- **Action map**: priority, track, gale swath and cancelled ferries
- **Risk vs access**, with the action tier for every province
- **Provinces**: expected cases for 4 weeks with a 90% range, people cut off (both methods), drivers
- **Municipalities**, **Outbreak graph**, **Forecast skill**, **Leptospirosis**, **Validation**, **Data & caveats**, and a built-in **Usability test**
- 98 scenarios pre-computed: answers in seconds

PUTTING IT IN THE LOOP

1
STORM WATCH

A PAGASA tropical cyclone bulletin goes in: its forecast track and shifted tracks for the cone of uncertainty.

2
PRE-LANDFALL BULLETIN

A ranked province list with action tiers to DOH regional offices, provincial health offices and DRRMOs, 48-72 h ahead.

3
POST-LANDFALL UPDATE

Days 1-7: Sentinel-1 flood maps replace the hazard estimate; ferry and road status update who is cut off.

4
LEARN EVERY WEEK

Weeks 1-8: each PIDSR report is scored against the forecast; R is recalibrated and tiers are reviewed with users.

The output is an **action**, not a chart: monitor sentinel sites, pre-position doxycycline and vector control, or send mobile clinics **before** the roads and ferries close.

1,641
municipalities with access loss for every scenario

98
storm scenarios ready on the dashboard

100%
open data and open code; no maps need a tile server

1
command rebuilds everything when new data arrive

The novelty is the join: **storm-specific** disease risk × **storm-specific** loss of access, ferries included, in one ranking a health officer can act on.

SIGNAL 20 / SOURCES

SOURCES: APA 7TH EDITION

Ansari, A. F., Shchur, O., Küken, J., Auer, A., Han, B., Mercado, P., Rangapuram, S. S., Shen, H., Stella, L., Shen, X., Turkmen, C., Bohlke-Schneider, M., Erickson, N., Kapoor, S., Li, Z., Wang, B., Williams, A. R., Ahmed, A., Weber, S., … Wang, Y. (2025). *Chronos-2: From univariate to universal forecasting*. arXiv. https://arxiv.org/abs/2510.15821

Balbuena, A., Herbosa, T., & Gundran, C. P. (2010). Cases of leptospirosis after Typhoon Ondoy (international name: Ketsana) seen at the emergency department: The Philippine General Hospital experience. *WIT Transactions on Information and Communication Technologies, 43*, 605–612. https://doi.org/10.2495/RISK100501

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological), 57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment, 2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Bracher, J., Ray, E. L., Gneiting, T., & Reich, N. G. (2021). Evaluating epidemic forecasts in an interval format. *PLoS Computational Biology, 17*(2), e1008618. https://doi.org/10.1371/journal.pcbi.1008618

Brooke, J. (1996). SUS: A "quick and dirty" usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), *Usability evaluation in industry* (pp. 189–194). Taylor & Francis.

Buczak, A. L., Baugher, B., Babin, S. M., Ramac-Thomas, L. C., Guven, E., Elbert, Y., Koshute, P. T., Velasco, J. M. S., Roque, V. G., Jr., Tayag, E. A., Yoon, I.-K., & Lewis, S. H. (2014). Prediction of high incidence of dengue in the Philippines. *PLoS Neglected Tropical Diseases, 8*(4), e2771. https://doi.org/10.1371/journal.pntd.0002771

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). https://doi.org/10.1145/2939672.2939785

Copernicus Emergency Management Service. (2018). *EMSR312: Typhoon Mangkhut in the Philippines* [Rapid mapping activation]. European Commission. https://mapping.emergency.copernicus.eu/activations/EMSR312/

Department of Health Epidemiology Bureau. (2021). *DOH-Epi dengue data 2016–2021* [Data set]. Humanitarian Data Exchange. https://data.humdata.org/dataset/ac63a95e-7296-42fb-802b-7f7541c73e45

Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A., & Michaelsen, J. (2015). The climate hazards infrared precipitation with stations—A new environmental record for monitoring extremes. *Scientific Data, 2*, 150066. https://doi.org/10.1038/sdata.2015.66

Gibbs, I., & Candès, E. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems, 34*, 1660–1672.

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment, 202*, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

Hoo, S. B., Müller, S., Salinas, D., & Hutter, F. (2025). *From tables to time: Extending TabPFN-v2 to time series forecasting*. arXiv. https://arxiv.org/abs/2501.02945

Huffman, G. J., Stocker, E. F., Bolvin, D. T., Nelkin, E. J., & Tan, J. (2023). *GPM IMERG final precipitation L3 half hourly 0.1 degree x 0.1 degree V07* [Data set]. NASA Goddard Earth Sciences Data and Information Services Center. https://doi.org/10.5067/GPM/IMERG/3B-HH/07

Knapp, K. R., Kruk, M. C., Levinson, D. H., Diamond, H. J., & Neumann, C. J. (2010). The International Best Track Archive for Climate Stewardship (IBTrACS): Unifying tropical cyclone data. *Bulletin of the American Meteorological Society, 91*(3), 363–376. https://doi.org/10.1175/2009BAMS2755.1

Lehner, B., & Grill, G. (2013). Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems. *Hydrological Processes, 27*(15), 2171–2186. https://doi.org/10.1002/hyp.9740

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*, 4765–4774.

Matsushita, N., Ng, C. F. S., Kim, Y., Suzuki, M., Saito, N., Ariyoshi, K., Salva, E. P., Dimaano, E. M., Villarama, J. B., Go, W. S., & Hashizume, M. (2018). The non-linear and lagged short-term relationship between rainfall and leptospirosis and the intermediate role of floods in the Philippines. *PLoS Neglected Tropical Diseases, 12*(4), e0006331. https://doi.org/10.1371/journal.pntd.0006331

OpenStreetMap contributors. (2026). *OpenStreetMap data extract: Philippines* [Data set]. Geofabrik. https://download.geofabrik.de/asia/philippines.html

Page, L., Brin, S., Motwani, R., & Winograd, T. (1999). *The PageRank citation ranking: Bringing order to the web* (Technical Report 1999-66). Stanford InfoLab.

Panagopoulos, G., Nikolentzos, G., & Vazirgiannis, M. (2021). Transfer graph neural networks for pandemic forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence, 35*(6), 4838–4845. https://doi.org/10.1609/aaai.v35i6.16616

Petricola, S., Reinmuth, M., Lautenbach, S., Hatfield, C., & Zipf, A. (2022). Assessing road criticality and loss of healthcare accessibility during floods: The case of Cyclone Idai, Mozambique 2019. *International Journal of Health Geographics, 21*, 14. https://doi.org/10.1186/s12942-022-00315-2

Philippine Statistics Authority. (2021). *Philippines: 2020 census of population and housing, population by municipality* [Data set]. Humanitarian Data Exchange. https://data.humdata.org/dataset/c1edf0c5-58c8-4d39-b566-937b1ad7907f

Philippine Statistics Authority & National Mapping and Resource Information Authority. (2025). *Philippines: Subnational administrative boundaries (COD-AB)* [Data set]. Humanitarian Data Exchange. https://data.humdata.org/dataset/caf116df-f984-4deb-85ca-41b349d3f313

Runge, J. (2020). Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets. In *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence* (PMLR 124, pp. 1388–1397). https://arxiv.org/abs/2003.03685

Runge, J., Nowack, P., Kretschmer, M., Flaxman, S., & Sejdinovic, D. (2019). Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances, 5*(11), eaau4996. https://doi.org/10.1126/sciadv.aau4996

Tariverdi, M., Nunez-del-Prado, M., Leonova, N., & Rentschler, J. (2023). Measuring accessibility to public services and infrastructure criticality for disasters risk management. *Scientific Reports, 13*, 1569. https://doi.org/10.1038/s41598-023-28460-z

Tatem, A. J. (2017). WorldPop, open data for spatial demography. *Scientific Data, 4*, 170004. https://doi.org/10.1038/sdata.2017.4

Tellman, B., Sullivan, J. A., Kuhn, C., Kettner, A. J., Doyle, C. S., Brakenridge, G. R., Erickson, T. A., & Slayback, D. A. (2021). Satellite imaging reveals increased proportion of population exposed to floods. *Nature, 596*, 80–86. https://doi.org/10.1038/s41586-021-03695-w

Tiecke, T. G., Liu, X., Zhang, A., Gros, A., Li, N., Yetman, G., Kilic, T., Murray, S., Blankespoor, B., Prydz, E. B., & Dang, H.-A. H. (2017). *Mapping the world population one building at a time*. arXiv. https://arxiv.org/abs/1712.05839

UN-SPIDER. (n.d.). *Recommended practice: Flood mapping and damage assessment using Sentinel-1 SAR data in Google Earth Engine*. United Nations Office for Outer Space Affairs. https://www.un-spider.org/advisory-support/recommended-practices/recommended-practice-google-earth-engine-flood-mapping

van Panhuis, W. G., Choisy, M., Xiong, X., Chok, N. S., Akarasewi, P., Iamsirithaworn, S., Lam, S. K., Chong, C. K., Lam, F. C., Phommasak, B., Vongphrachanh, P., Bouaphanh, K., Rekol, H., Hien, N. T., Thai, P. Q., Duong, T. N., Chuang, J.-H., Liu, Y.-L., Ng, L.-C., … Burke, D. S. (2015). Region-wide synchrony and traveling waves of dengue across eight countries in Southeast Asia. *Proceedings of the National Academy of Sciences, 112*(42), 13069–13074. https://doi.org/10.1073/pnas.1501375112

Weiss, D. J., Nelson, A., Vargas-Ruiz, C. A., Gligorić, K., Bavadekar, S., Gabrilovich, E., Bertozzi-Villa, A., Rozier, J., Gibson, H. S., Shekel, T., Kamath, C., Lieber, A., Schulman, K., Shao, Y., Qarkaxhija, V., Nandi, A. K., Keddie, S. H., Rumisha, S., Amratia, P., … Gething, P. W. (2020). Global maps of travel time to healthcare facilities. *Nature Medicine, 26*, 1835–1838. https://doi.org/10.1038/s41591-020-0931-3

Zadrozny, B., & Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 694–699). https://doi.org/10.1145/775047.775151